a Identification of bistable ecosystems

Climtic factor acts as a stressor

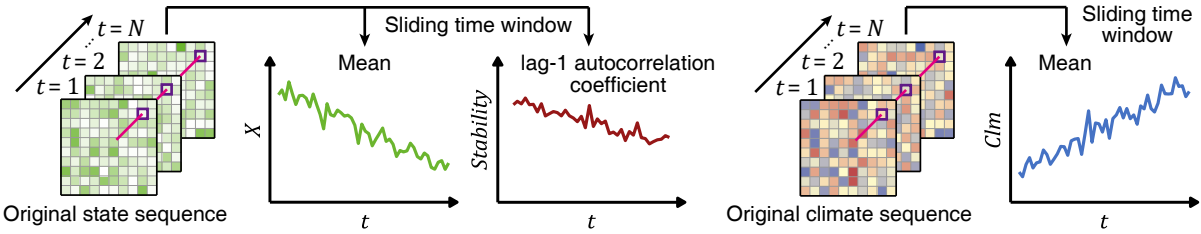
$$\left(\frac{dX}{dt}\right)_t = r\frac{X_t}{K}\left(K' - \frac{X_t}{K}\right)\left(\frac{X_t}{K} - c\right) + v - \alpha Clm_t X_t$$

$$Stability_t = \frac{3r}{K^3}X_t^2 - \frac{2rK' + 2rc}{K^2}X_t + \frac{rcK'}{K} + \alpha Clm_t$$

Climtic factor acts as a facilitator

$$\left(\frac{dX}{dt}\right)_t = r\frac{X_t}{K}\left(K' - \frac{X_t}{K}\right)\left(\frac{X_t}{K} - c\right) + v + \alpha Clm_t X_t$$

$$Stability_t = \frac{3r}{K^3}X_t^2 - \frac{2rK' + 2rc}{K^2}X_t + \frac{rcK'}{K} - \alpha Clm_t$$

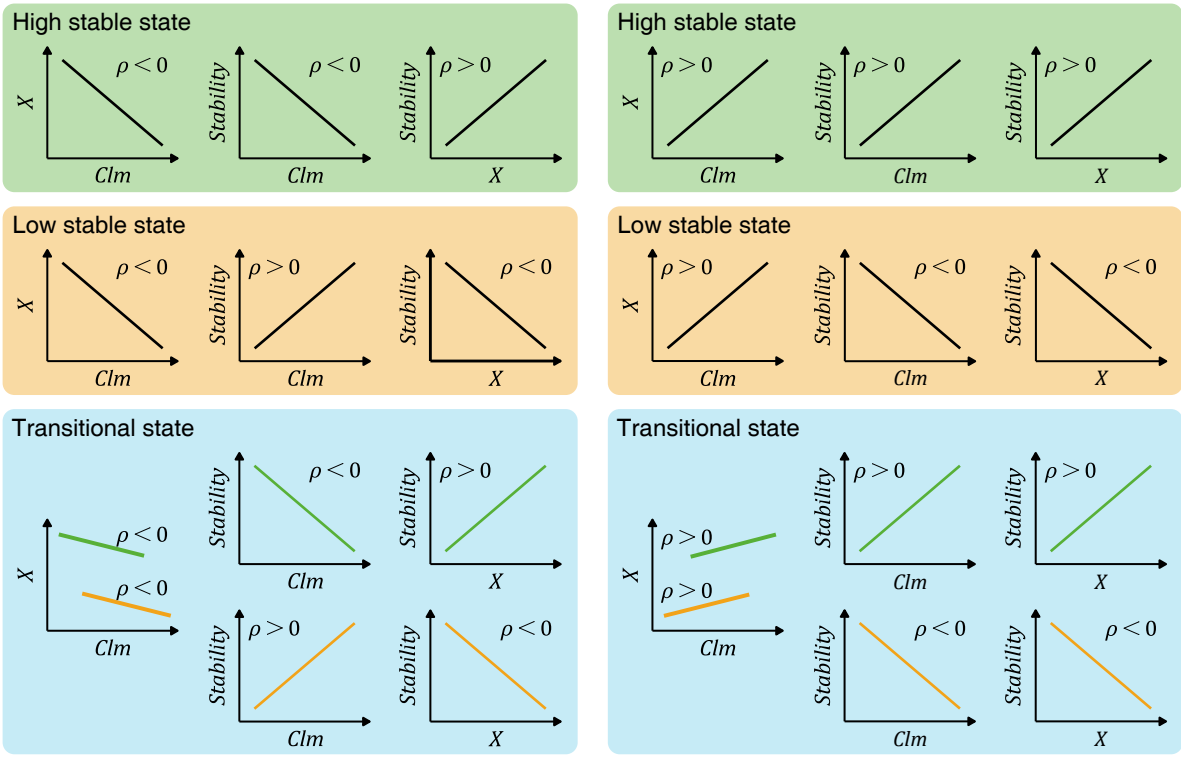
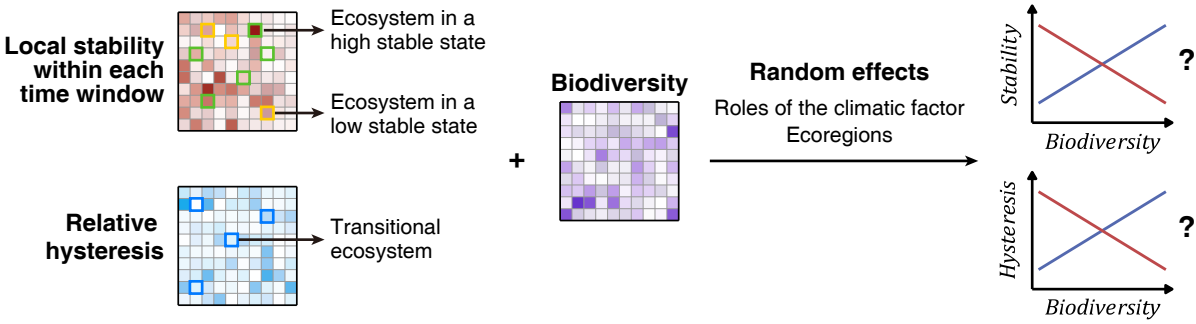


$$Stability_t = \lambda_1 X_t^2 + \lambda_2 X_t + \lambda_3 + \lambda_4 Clm_t$$

$\lambda_1 > 0,\ \lambda_2 < 0,\ \lambda_3 > 0,\ \lambda_4 > 0;\ P < 0.05$

$\lambda_1 > 0,\ \lambda_2 < 0,\ \lambda_3 > 0,\ \lambda_4 < 0;\ P < 0.05$

b Determination of developmental stages



c Evaluation of biodiversity effects